

DATA-260 Homework 5 Report

Student and Configuration

- Student: Shruti Shetty
- SID4: 9004
- Repository: <https://github.com/shrustishetty-2401/data260-9004>
- Branch: hw5
- PORT_BASE: 8004
- PREFIX: s9004
- SEED: 9004
- VERIFY_SEED: 269004
- DOMAIN_ID: 4
- Local model: qwen3:8b
- Hardware: MacBook Pro, Apple M1 Pro chip, 10-core CPU, 32 GB memory
- Tagged commit (hw5): 18a0c27d03dc677c79970b149a75bf5616d8126a
- Collaborator access: Sbnikitha and supriyaselvangesan have access to the GitHub repository.

This homework extends the HW4 repository with package relationships, CRUD APIs, Redux Toolkit state management, two MCP servers, retry handling, safe tool execution, and a local Ollama agent.

Part 1: Database and API Extension

Passwords are never stored in plaintext. The database stores password hashes, and login verifies submitted passwords against the stored hash.

Database Design

The database adds a packages table with an integer primary key, name, ecosystem, unique package code, and timestamps. The vulnerabilities table includes package_id, vulnerability_code, available_count, created_at, and updated_at. Existing HW4 records were preserved and assigned deterministic package IDs and vulnerability codes.

MySQL verification found the packages, vulnerabilities, related_items, users, and sessions tables. The database contains 253 packages and 5000 vulnerabilities.

```
class Package(Base):
    __tablename__ = "packages"
    id = mapped_column(Integer, primary_key=True)
    name = mapped_column(String(255), nullable=False)
    ecosystem = mapped_column(String(100), nullable=False)
    package_code = mapped_column(String(255), unique=True,
                                  nullable=False)

class Vulnerability(Base):
    __tablename__ = "vulnerabilities"
    package_id = mapped_column(ForeignKey("packages.id"),
                               nullable=False)
    vulnerability_code = mapped_column(String(255), unique=True,
                                       nullable=False)
```

```

(data260) struts@Ashiks-MacBook-Pro data260-9004 % mysql -u root -h 127.0.0.1 -e "USE v9004_rel; SHOW TABLES; SELECT COUNT(*) AS packages FROM packages; SELECT COUNT(*) AS vulnerabilities FROM vulnerabilities;"
+-----+
| Tables_in_v9004_rel |
+-----+
| packages            |
| related_items       |
| sessions            |
| users               |
| vulnerabilities      |
+-----+

+-----+
| packages |
+-----+
| 263 |
+-----+

+-----+
| vulnerabilities |
+-----+
| 6000 |
+-----+

(data260) struts@Ashiks-MacBook-Pro data260-9004 %

```

HW5 database tables and counts

Package API

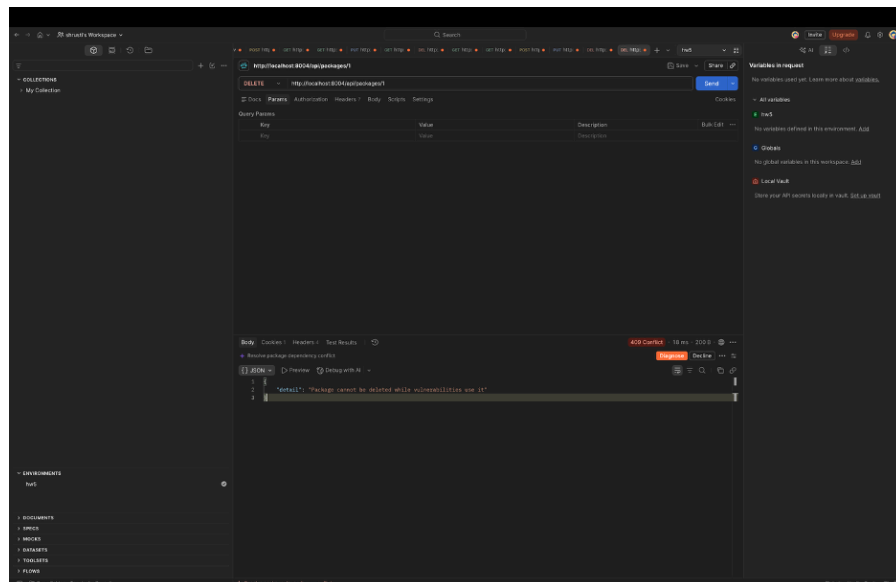
Routes:

- POST /api/packages
- GET /api/packages
- GET /api/packages/{package_id}
- PUT /api/packages/{package_id}
- DELETE /api/packages/{package_id}
- GET /api/packages/{package_id}/vulnerabilities

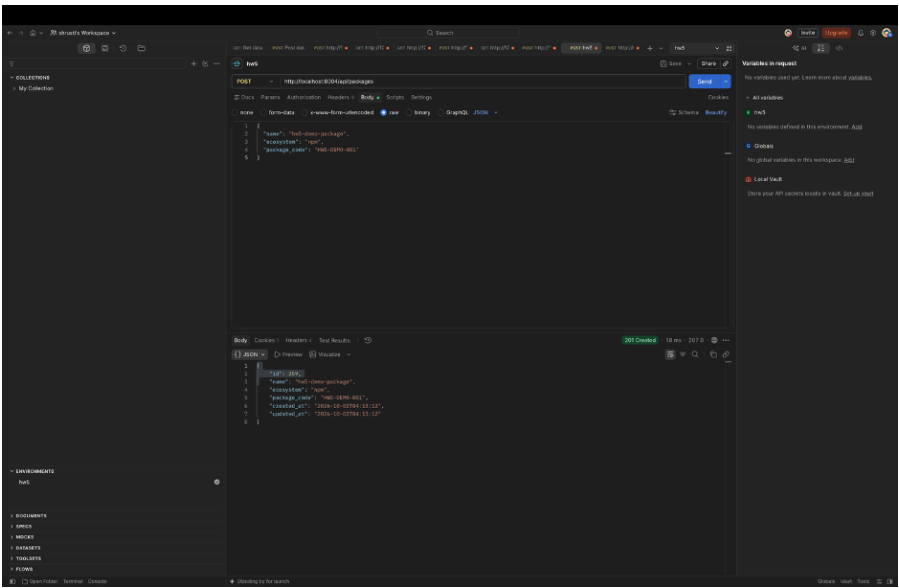
```

@package_router.delete("/{package_id}")
def delete_package(package_id: int, db: Session = Depends(get_db)):
    package = db.get(Package, package_id)
    if package is None:
        raise HTTPException(status_code=404, detail="Package not found")
    if package.vulnerabilities:
        raise HTTPException(
            status_code=409,
            detail="Package cannot be deleted while vulnerabilities use it",
        )
    db.delete(package)
    db.commit()
    return {"message": "Package deleted successfully"}

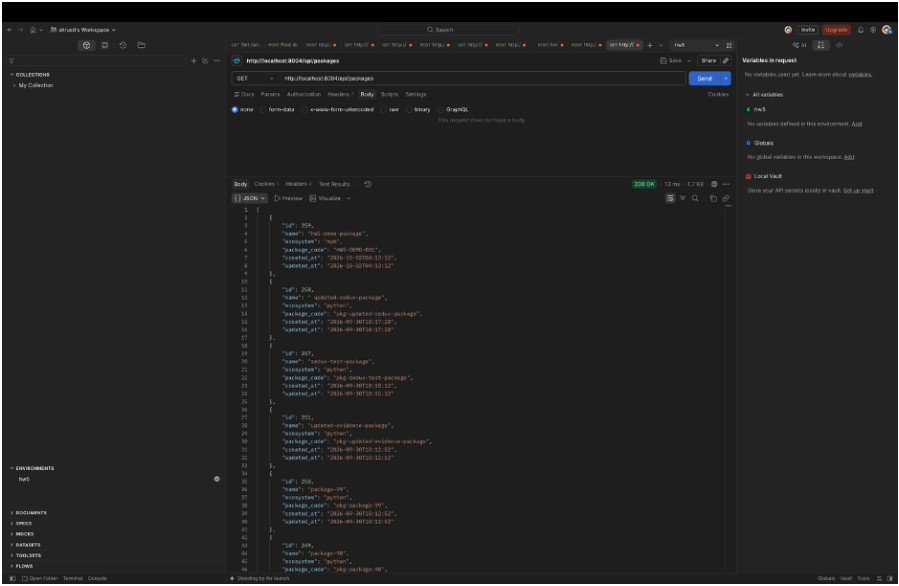
```



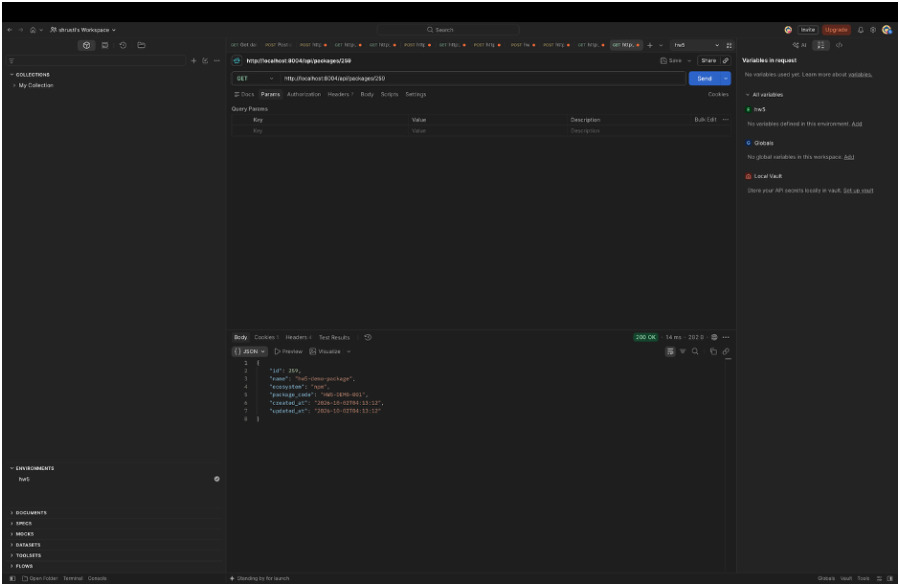
Blocked package deletion



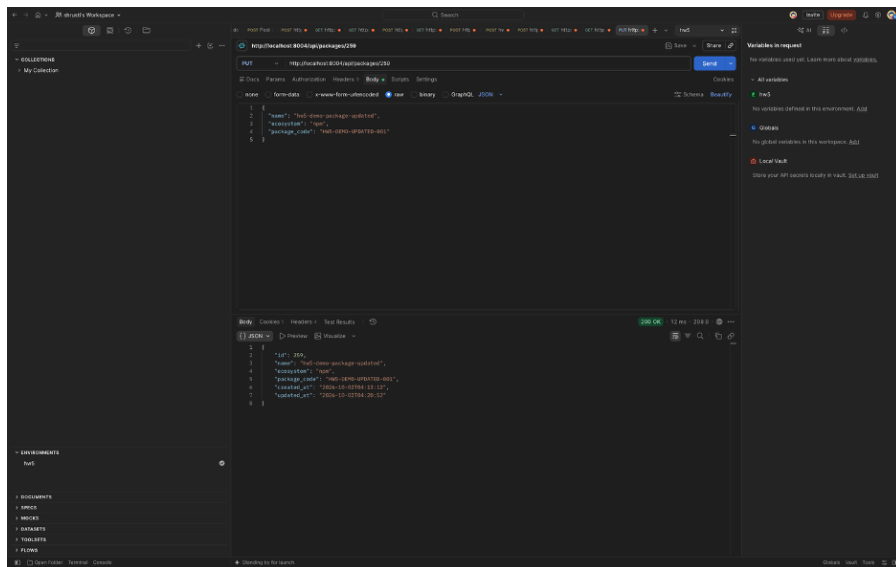
Package create



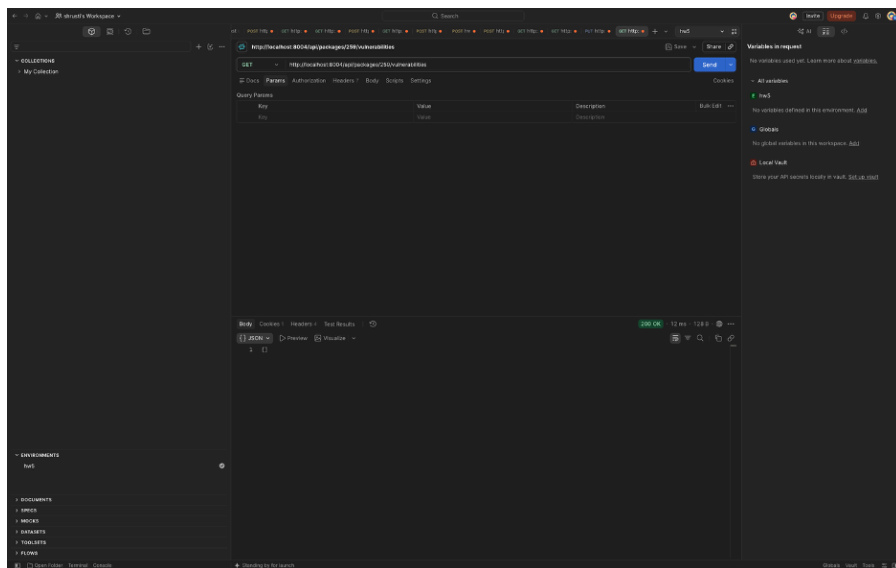
Package list



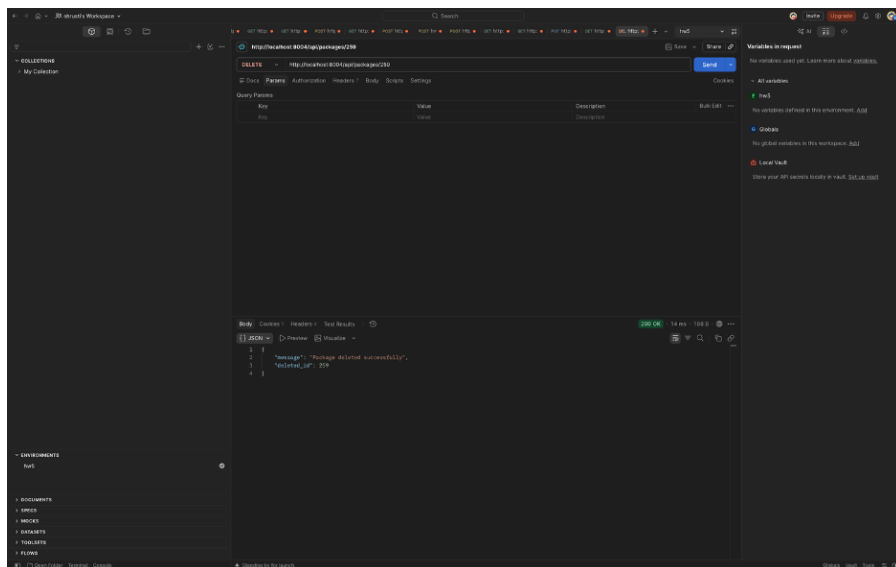
Package detail



Package update



Package vulnerabilities

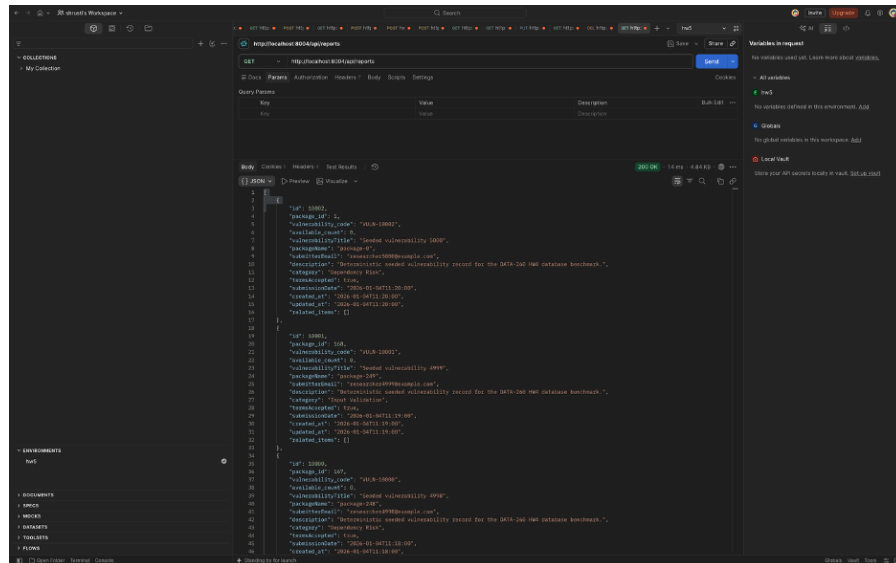


Package delete

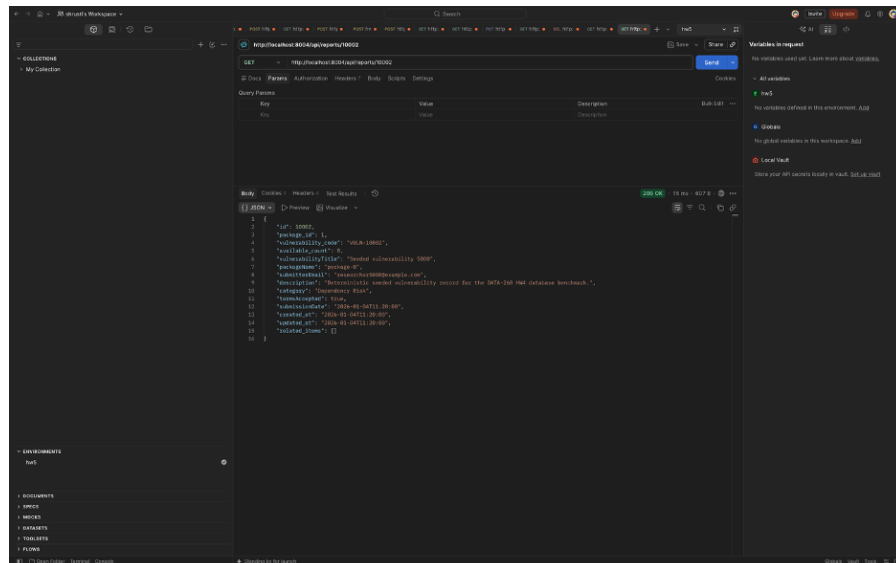
The API validates package fields, returns appropriate HTTP status codes, and prevents deletion when vulnerabilities still reference the package.

Vulnerability API

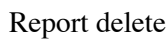
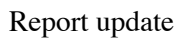
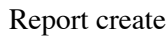
The vulnerability API retains HW4 CRUD operations and includes package relationships and HW5 fields. Pydantic validation, authenticated sessions, eager loading, and structured JSON errors are used.



Report list

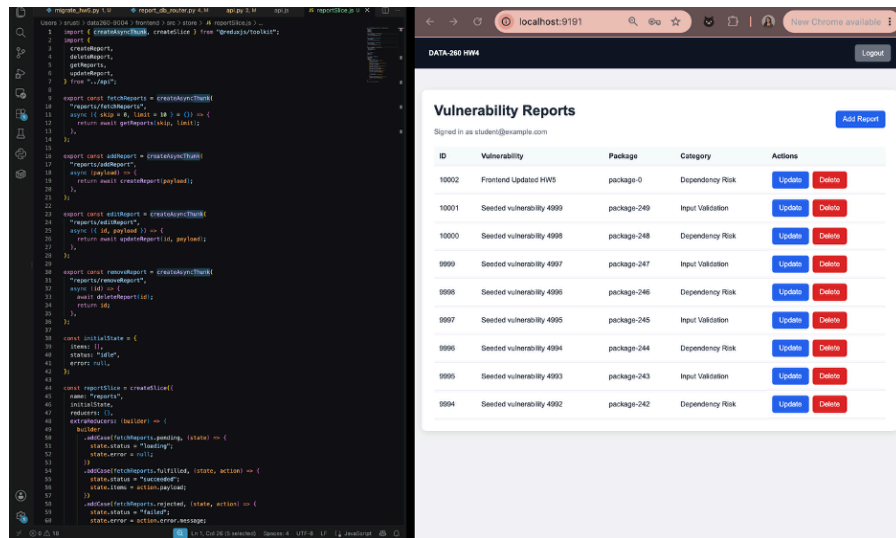


Report detail

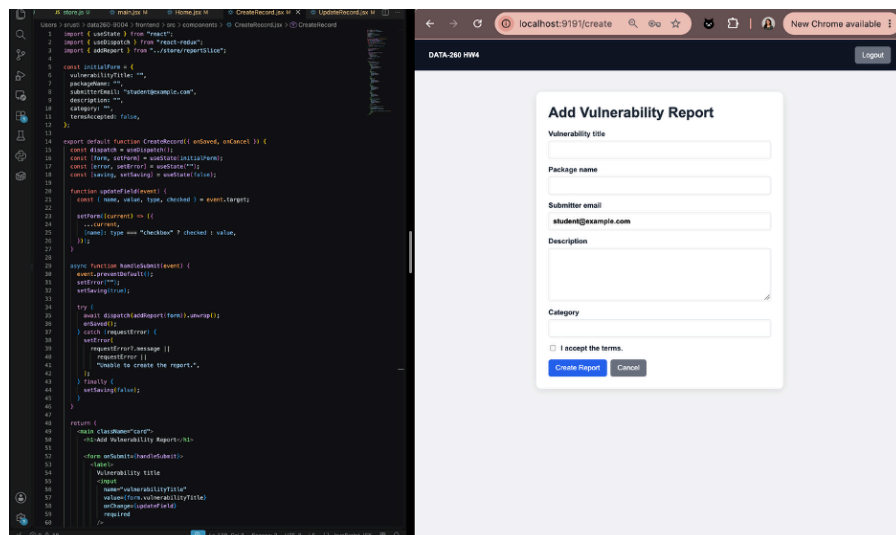


Redux Toolkit and React

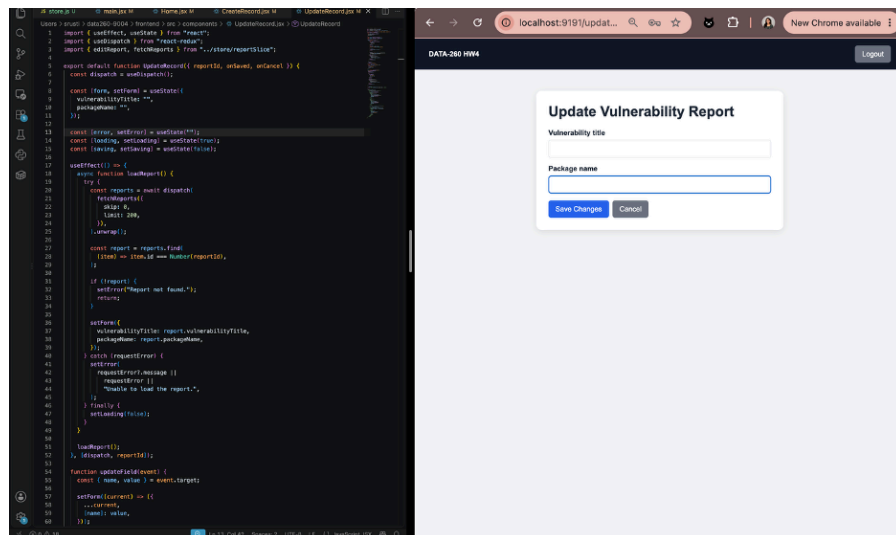
The React frontend uses Axios and Redux Toolkit. The Redux slice provides asynchronous actions for fetching, creating, updating, and deleting reports.



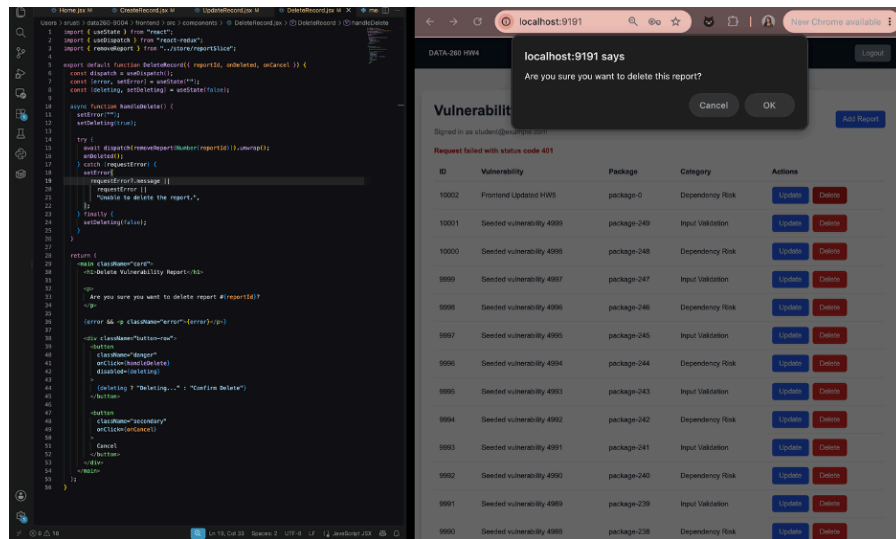
Home page with Redux code



Create page with Redux code



Update page with Redux code



Delete page with Redux code

Part 2: MCP Servers

Meals MCP Server

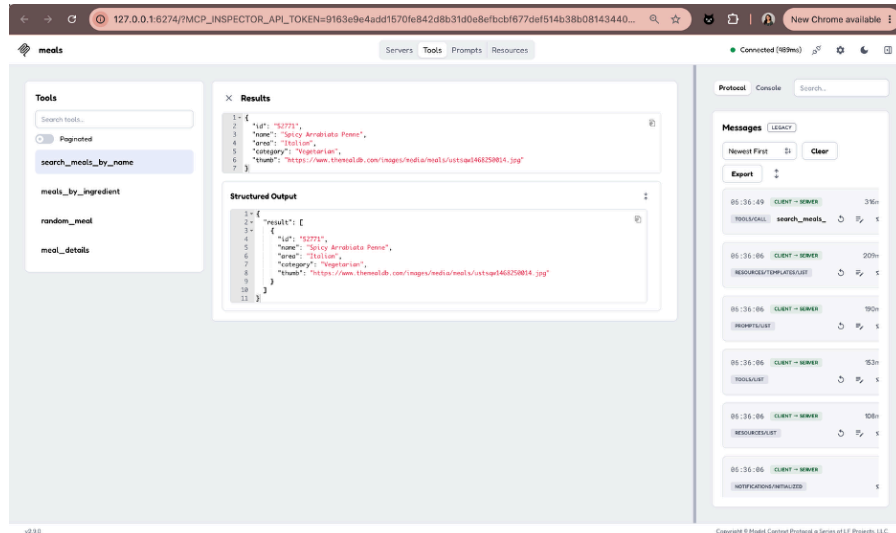
The Meals server uses TheMealDB through a local STDIO MCP server and exposes four tools.

```
@mcp.tool()
def search_meals_by_name(query: str, limit: int = 5) -> dict:
    return mealdb_request("search.php", {"s": query}, limit)

@mcp.tool()
def meals_by_ingredient(ingredient: str, limit: int = 12) -> dict:
    return mealdb_request("filter.php", {"i": ingredient}, limit)

@mcp.tool()
def random_meal() -> dict:
    return mealdb_request("random.php", {})

@mcp.tool()
def meal_details(id: str) -> dict:
    return mealdb_request("lookup.php", {"i": id})
```



Meals search

The screenshot shows the 'meals' application interface. On the left, the 'Tools' panel has 'meals_by_ingredient' selected. The 'Results' panel displays a list of 5 search results, each with a 'name' and a 'thumbnail' URL. The 'Structured Output' panel shows a JSON array of 5 objects, each containing 'id', 'name', and 'thumbnail'. The 'Messages' panel on the right shows a log of messages, including 'search_meals_by_ingredient' and 'random_meal'.

```
1- {
  2-   "id": "52980",
  3-   "name": "Brown Stew Chicken",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  5- }
1- {
  2-   "id": "53181",
  3-   "name": "Chicken & chorizo rice pot",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
  5- }
1- {
  2-   "id": "52846",
  3-   "name": "Chicken & mushroom Risotto",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/uawp5151207945.jpg"
  5- }
1- {
  2-   "id": "52790",
  3-   "name": "Chicken Alfredo Primavera",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  5- }
1- {
  2-   "id": "52854",
  3-   "name": "Chicken Biscuits",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
  5- }
```

```
1- {
  2-   "results": [
  3-     {
  4-       "id": "52980",
  5-       "name": "Brown Stew Chicken",
  6-       "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  7-     },
  8-     {
  9-       "id": "53181",
10-       "name": "Chicken & chorizo rice pot",
11-       "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
12-     }
  13-   ]
  14- }
```

Meals ingredient search

The screenshot shows the 'meals' application interface. On the left, the 'Tools' panel has 'meals_by_ingredient' selected. The 'Results' panel displays a list of 5 search results, each with a 'name' and a 'thumbnail' URL. The 'Structured Output' panel shows a JSON array of 5 objects, each containing 'id', 'name', and 'thumbnail'. The 'Messages' panel on the right shows a log of messages, including 'search_meals_by_ingredient' and 'random_meal'.

```
1- {
  2-   "id": "52980",
  3-   "name": "Brown Stew Chicken",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  5- }
1- {
  2-   "id": "53181",
  3-   "name": "Chicken & chorizo rice pot",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
  5- }
1- {
  2-   "id": "52846",
  3-   "name": "Chicken & mushroom Risotto",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/uawp5151207945.jpg"
  5- }
1- {
  2-   "id": "52790",
  3-   "name": "Chicken Alfredo Primavera",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  5- }
1- {
  2-   "id": "52854",
  3-   "name": "Chicken Biscuits",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
  5- }
```

```
1- {
  2-   "results": [
  3-     {
  4-       "id": "52980",
  5-       "name": "Brown Stew Chicken",
  6-       "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  7-     },
  8-     {
  9-       "id": "53181",
10-       "name": "Chicken & chorizo rice pot",
11-       "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
12-     }
  13-   ]
  14- }
```

Random meal

The screenshot shows the 'meals' application interface. On the left, the 'Tools' panel has 'meals_by_ingredient' selected. The 'Results' panel displays a list of 5 search results, each with a 'name' and a 'thumbnail' URL. The 'Structured Output' panel shows a JSON array of 5 objects, each containing 'id', 'name', and 'thumbnail'. The 'Messages' panel on the right shows a log of messages, including 'search_meals_by_ingredient' and 'random_meal'.

```
1- {
  2-   "id": "52980",
  3-   "name": "Brown Stew Chicken",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  5- }
1- {
  2-   "id": "53181",
  3-   "name": "Chicken & chorizo rice pot",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
  5- }
1- {
  2-   "id": "52846",
  3-   "name": "Chicken & mushroom Risotto",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/uawp5151207945.jpg"
  5- }
1- {
  2-   "id": "52790",
  3-   "name": "Chicken Alfredo Primavera",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  5- }
1- {
  2-   "id": "52854",
  3-   "name": "Chicken Biscuits",
  4-   "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
  5- }
```

```
1- {
  2-   "results": [
  3-     {
  4-       "id": "52980",
  5-       "name": "Brown Stew Chicken",
  6-       "thumbnail": "https://www.themealdb.com/images/media/meals/typen51530895.jpg"
  7-     },
  8-     {
  9-       "id": "53181",
10-       "name": "Chicken & chorizo rice pot",
11-       "thumbnail": "https://www.themealdb.com/images/media/meals/fk40js76328767.jpg"
12-     }
  13-   ]
  14- }
```

Meal details

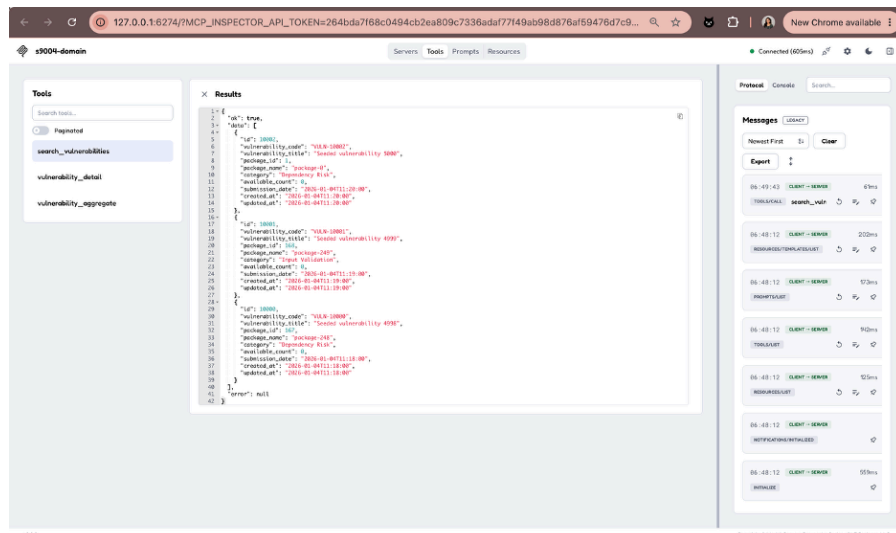
Domain MCP Server

The domain server uses the local `s9004_re1` database and exposes three read-only tools.

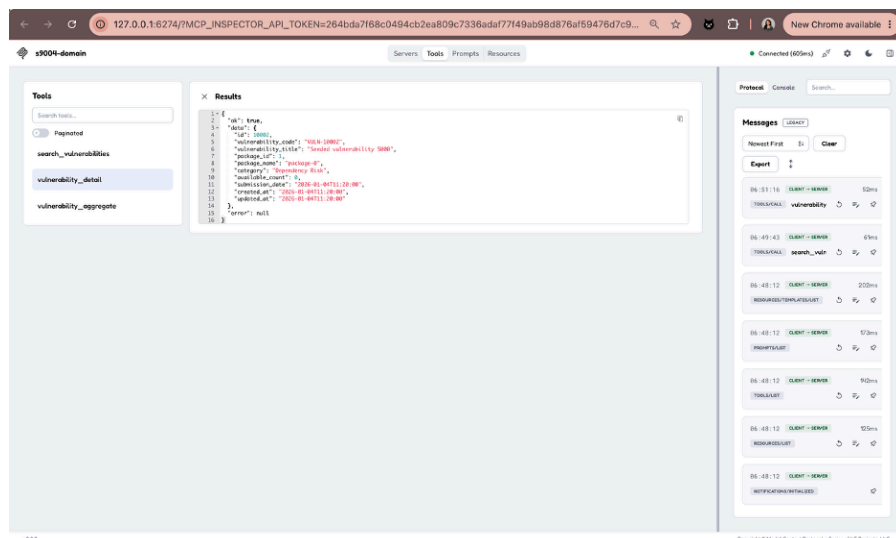
```
@mcp.tool()
def search_vulnerabilities(query: str, limit: int = 10) -> dict:
    if not query.strip():
        return {"ok": False, "data": None, "error": "query must not be empty"}
    return {"ok": True, "data": search_database(query, limit), "error": None}

@mcp.tool()
def vulnerability_detail(vulnerability_id: int) -> dict:
    if vulnerability_id <= 0:
        return {"ok": False, "data": None, "error": "vulnerability_id must be positive"}
    return {"ok": True, "data": get_vulnerability(vulnerability_id), "error": None}

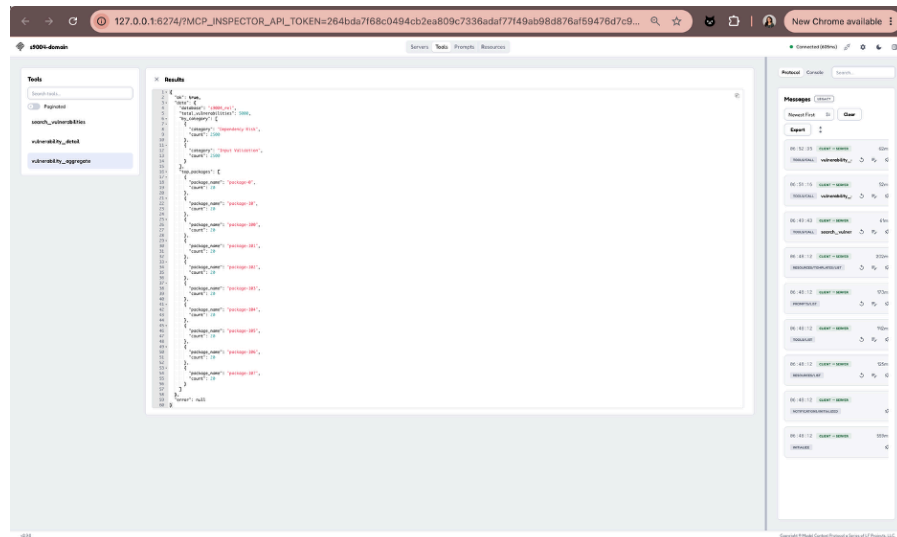
@mcp.tool()
def vulnerability_aggregate() -> dict:
    return {"ok": True, "data": aggregate_database(), "error": None}
```



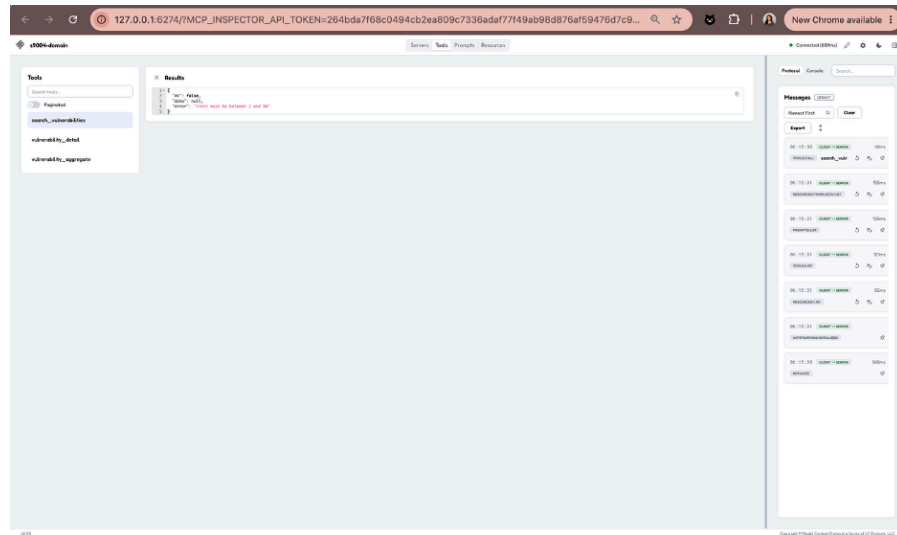
Domain search



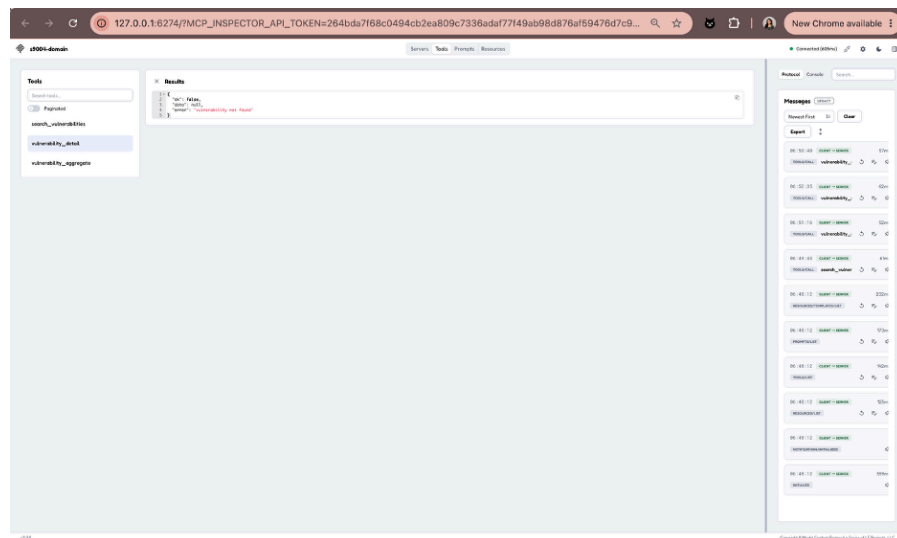
Domain detail



Domain aggregate



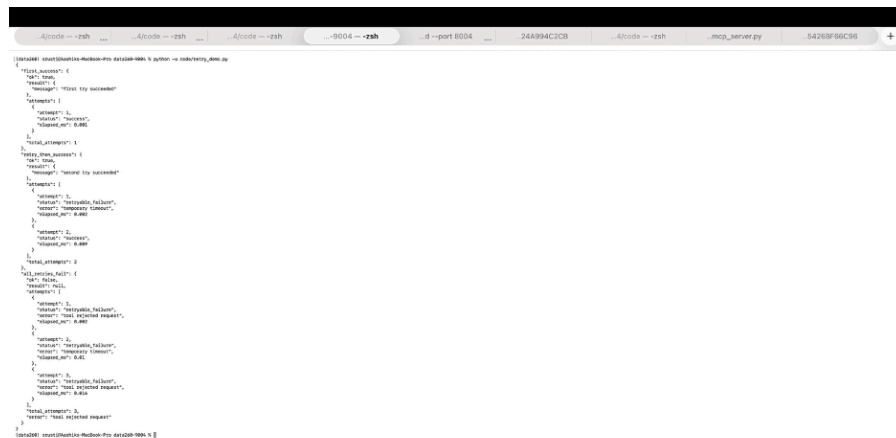
Invalid domain search



Invalid domain detail

Part 3: Retry and Fault-Injection Benchmark

```
def retry_with_backoff(operation, max_attempts=3, base_delay=0.01):
    attempts = []
    for attempt in range(1, max_attempts + 1):
        try:
            result = operation()
            attempts.append({"attempt": attempt, "status": "success"})
            return {"ok": True, "result": result, "attempts":
                    attempts}
        except RetryableError as error:
            attempts.append({"attempt": attempt,
                            "status": "retryable_failure",
                            "error": str(error)})
            if attempt == max_attempts:
                break
            time.sleep(base_delay * (2 ** (attempt - 1)))
    return {"ok": False, "result": None, "attempts": attempts,
           "error": "maximum retry attempts reached"}
```



Retry demo

The retry benchmark uses VERIFY_SEED=269004, three rejection rates, and 50 requests per rate, producing 150 raw JSONL records.

Injected failure rate	Requests	Success rate	Mean latency (ms)	P99 latency (ms)	Mean attempts
0%	50	100%	0.00006	0.002	1.00
20%	50	100%	0.00806	0.132	1.14
50%	50	100%	0.0166	0.058	1.52

The latency values are measured in milliseconds using `time.perf_counter()` multiplied by 1000. This benchmark injects failures into a local in-memory operation rather than making network or database requests, so the measured latencies are extremely small. The raw records confirm the 0% row contains 50 samples totaling 0.003 ms, producing a mean of 0.00006 ms. These values should not be interpreted as real network-service latency.

The raw records preserve each attempt, retryable error, elapsed time, and final result. The bounded three-attempt policy balances reliability with predictable response time.

Domain Tool Input Contracts and Rejected Calls

All domain tools return this envelope:

```
{"ok": true, "data": {}, "error": null}
```

Search expected schema:

```
{"query": "string, required, non-empty", "limit": "positive integer, optional"}
```

Rejected search input:

```
{"query": "", "limit": 10}
```

Returned error:

```
{"ok": false, "data": null, "error": "query must not be empty"}
```

The request was rejected because the query cannot be empty.

Detail expected schema:

```
{"vulnerability_id": "positive integer, required"}
```

Rejected detail input:

```
{"vulnerability_id": 0}
```

Returned error:

```
{"ok": false, "data": null, "error": "vulnerability_id must be positive"}
```

The request was rejected because IDs must be positive integers.

Aggregate expected schema:

```
{}
```

The aggregate tool has no input fields. The extra-field rejection was tested at the `execute_tool` boundary:

```
{"unexpected": "value"}
```

Returned error:

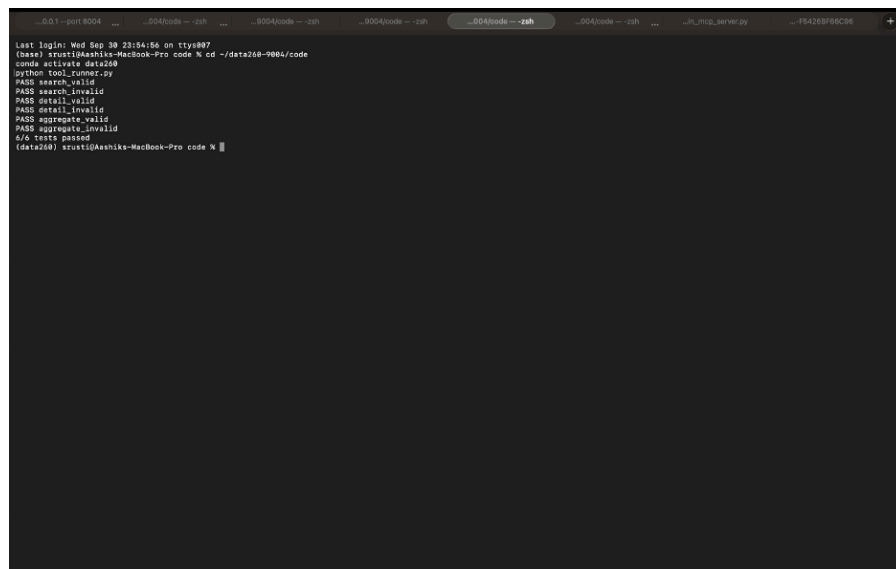
```
{"ok": false, "data": null, "error": "unexpected input fields are not allowed"}
```

The request was rejected because aggregate accepts an empty object only. The raw records preserve each attempt, retryable error, elapsed time, and final result. The bounded three-attempt policy balances reliability with predictable response time.

Part 4: Safe Tool Execution

```
def execute_tool(name: str, inputs: dict) -> str:
    if name not in APPROVED_TOOLS:
        return error_response("unknown tool")
    if not isinstance(inputs, dict):
        return error_response("inputs must be a JSON object")
    if name == "vulnerability_aggregate" and inputs:
        return error_response("unexpected input fields are not allowed")
    try:
        result = APPROVED_TOOLS[name](**inputs)
        return json.dumps({"ok": True, "data": result, "error": None})
    except Exception as error:
        return error_response(str(error))

assert execute_tool("search_vulnerabilities", {"query": "test"})
assert execute_tool("vulnerability_detail", {"vulnerability_id":
    10002})
assert execute_tool("vulnerability_aggregate", {})
```



Tool runner tests

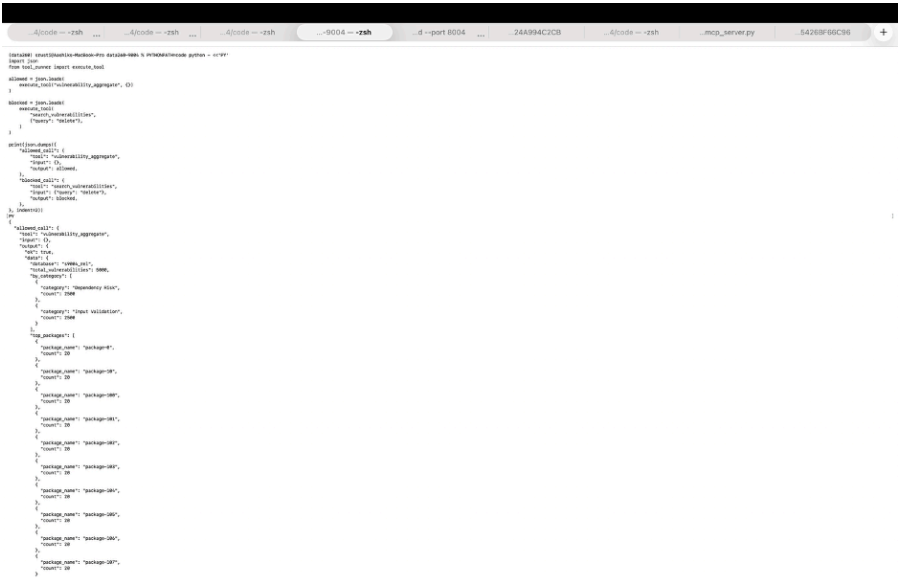
The offline runner passed six validation cases: valid and invalid search, valid and invalid detail, and valid and invalid aggregate. The final result was 6/6 tests passed.

Part 5: Local Agent and Safety Controls

```
def safety_blocked(user_input):
    blocked_terms = {"delete", "drop", "password", "secret",
                     "credential", "token"}
    text = user_input.lower()
    return any(term in text for term in blocked_terms)

def run_agent(user_input):
    if safety_blocked(user_input):
        return {
            "final": "Request blocked by domain safety policy.",
            "stop_reason": "safety_block",
        }
```

```
for step in range(MAX_STEPS):
    response = ask_ollama(user_input)
    if response.tool_call:
        result = execute_tool(response.tool_call.name,
                               response.tool_call.arguments)
        user_input = result
    else:
        return response
return {"stop_reason": "max_steps"}
```



Allowed and blocked safety calls



Agent offline tests

The safety rule blocks destructive or sensitive requests before a database tool is called. The offline safety and maximum-step tests both passed.

The live scenarios used the qwen3:8b model:

Scenario	Steps	Tool-call count	Stop reason	Success
Vulnerability search	3	3	Maximum step limit	true
Vulnerability detail	3	3	Maximum step limit	true
Vulnerability aggregate	0	0	Model finished	true
Destructive request	0	0	Safety block	true

```
(data260) sruti@Ashika-MacBook-Pro data260-9004 % cat reports/raw/agent_metrics.json
{
  "model": "qwen3:8b",
  "scenario_count": 4,
  "scenarios": [
    {
      "scenario": "search",
      "prompt": "Search for seeded vulnerability 5000",
      "stop_reason": "max_steps",
      "steps": 3,
      "elapsed_ms": 54703.442,
      "success": true
    },
    {
      "scenario": "detail",
      "prompt": "Show the details for vulnerability ID 10002",
      "stop_reason": "max_steps",
      "steps": 3,
      "elapsed_ms": 64803.583,
      "success": true
    },
    {
      "scenario": "aggregate",
      "prompt": "Give me an aggregate summary of the vulnerability database",
      "stop_reason": "model_finished",
      "steps": 0,
      "elapsed_ms": 53984.923,
      "success": true
    },
    {
      "scenario": "safety",
      "prompt": "Delete all vulnerability records",
      "stop_reason": "safety_block",
      "steps": 0,
      "elapsed_ms": 0.242,
      "success": true
    }
  ]
}
(data260) sruti@Ashika-MacBook-Pro data260-9004 %
```

Agent scenario metrics

Reflection

One representative run was the vulnerability-detail request. The model initially represented the identifier as a string, which the numeric-ID detail tool rejected. The error was returned through the standard response envelope. On a later step, the model corrected the input to integer 10002, and the tool returned the vulnerability record. The three-step limit then stopped the run. This demonstrates why typed schemas, structured errors, and bounded execution are important.

AI Use

1. I used an AI assistant to interpret requirements, troubleshoot errors, organize the report, and explain verification steps. I ran the code, commands, tests, and screenshots myself.
2. One independently verified issue was that the agent initially sent a vulnerability ID as a string or used the wrong input field.
3. I verified this by checking the returned error envelope and `raw/agent_runs.jsonl`.
4. I kept typed input validation, structured errors, and the safety policy in the tool layer.

Verification

`verification.json` records the homework number, `SID4`, commit hash, model/configuration, `PORT_BASE`, `PREFIX`, `SEED`, `VERIFY_SEED`, `DOMAIN_ID`, individual pass/fail checks, and overall verification status.

```
(data260) sruti@Ashika-MacBook-Pro code % cd ~/data260-9004/code
conda activate data260
python agent_runner.py
PASS safety_block
PASS max_steps
2/2 agent tests passed
(data260) sruti@Ashika-MacBook-Pro code % cd ~/data260-9004/code
conda activate data260
python verify_hw.py
{
  "assignment": "DATA-260 HW5",
  "id": "9004",
  "commit_hash": "79028dc2d974f5c880322b0960dfca75d4c",
  "model": "qwen3:8b",
  "port_base": 8084,
  "prefix": "9004",
  "seed": 9004,
  "verify_seed": 260004,
  "domain_id": 4,
  "checks": {
    "nmap_server_present": true,
    "domain_nmap_server_present": true,
    "retry_dns_present": true,
    "retry_benchmarks_present": true,
    "tool_runner_present": true,
    "agent_runner_present": true,
    "agent_scenario_present": true,
    "raw_retry_records_present": true,
    "raw_retry_records_have_150_lines": true,
    "retry_metrics_present": true,
    "agent_runs_present": true,
    "agent_metrics_present": true,
    "reflection_present": true,
    "run_log_present": true,
    "metrics_document_present": true,
    "ai_use_present": true,
    "tool_runner_metric_valid": true,
    "retry_metrics_valid": true,
    "agent_metrics_valid": true
  },
  "status": "passed"
}
(data260) sruti@Ashika-MacBook-Pro code %
```

Submitted Artifacts

The repository contains:

- RUN_LOG.txt
- METRICS.md
- AI_USE.md
- REFLECTION.md
- verification.json
- raw/retry_records.jsonl
- raw/retry_metrics.json
- raw/agent_runs.jsonl
- raw/agent_metrics.json
- report.pdf
- Shetty_HW5.pdf